haptic headband bom

Item	Category	Qty	Unit Price (£)	Total (£)
Compute & connectivity				
Adafruit HUZZAH32 ESP32 breakout board	Electronics	1	£13 00	£13 00
Sensing				
VL53L5CX 8x8 Time of Flight sensor breakout	Electronics	2	£19 50	£39 00
Adafruit BNO085 9-DOF IMU breakout	Electronics	1	£27 00	£27 00
Haptic output				
Vibrating mini motor disc (10mm coin)	Electronics	3	£1 70	£5 10
NPN transistor (e.g. 2N2222) - motor driver	Electronics	1	£1 70	£1 70
Flyback diode (1N4148 or similar)	Electronics	1	£1 50	£1 50
Gate/base resistor	Electronics	1	£6 00	£6 00
Power				
2000mAh 3.7V LiPo battery, JST-PH connector	Electronics	1	£10 00	£10 00
Adafruit Micro-Lipo charger, USB-C jack	Electronics	1	£5 70	£5 70
Breadboard-friendly SPDT slide switch	Electronics	1	£0 80	£0 80
Wiring & assembly				
Silicone hookup wire, assorted, per unit	Hardware	1	£0 80	£0 80
JST connectors / headers, assorted	Hardware	1	£7 80	£7 80
Custom PCB probs from pcb way	Hardware	1	£2 40	£2 40
Software / positioning				
Maps routing API	Software	1	£0 00	£0 00
Hyperlocation SDK license	Software	1	£0 00	£0 00
				£120 80

Notes	Link
Runs nav/obstacle logic, Bluetooth to phone	View product
8x8 zone ToF, ~45 deg horizontal FOV, I2C, 2cm-4m range	View product
On-chip sensor fusion, gives heading directly, STEMMA QT/I2C	View product
Front, left temple, right temple	View product
One per motor; generic part, source from any electronics supplier	View product
One per motor, protects driver transistor; generic part	View product
One per motor driver; generic part, buy as an assorted pack	View product
Built-in protection circuit; sized after current draw testing	View product
Charges the LiPo above over USB-C; matches design's USB-C port	View product
Power on/off	View product
2m per roll; buy 2-3 rolls in different colours for motor + IMU runs	View product
Covers battery, sensor, and motor connectors with plenty spare; crimping pliers recommended sep	View product
Mounts ESP32 + driver circuitry; 3-pack gives spares for mistakes/second build	IDK
Cost depends on provider/usage tier, not itemized here	-
GPS-denied positioning fallback; confirm licensing terms with vendor	View product